ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 7

TECHNOLOGY

PRACTICE EXAM 2 · MARKING GUIDELINE

MARKS: 75

Marking guideline for the Academic Central Grade 7 Technology practice exam 2.

Written by Academic Central for practice, following the CAPS curriculum for Grade 7. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Accept answers in the learner's own words and other correct examples. In calculations give method marks even if the final answer is wrong.

QUESTION 1

1 1.1 B ✓ 1.2 A ✓ 1.3 B ✓ 1.4 C ✓ 1.5 A ✓ 1.6 B ✓ 1.7 B ✓ 1.8 B ✓ 1.9 B ✓ 1.10 B ✓ [10]

QUESTION 2

2.1 (a) Beam bridge ✓ (b) Suspension bridge ✓ (c) Arch bridge ✓ (3)

2.2 (a) A strut is a part that is in compression (it is pushed) ✓; a tie is a part that is in tension (it is pulled) ✓. (3)
(b) Any ONE: the cables of a suspension bridge; the tie beam of a roof truss; guy ropes of a tent ✓

2.3 (a) Any TWO: make the base wider ✓; keep the heavy parts low (low centre of gravity) ✓; anchor it to the ground; add guy ropes or braces (4)
(b) The blocks are a counterweight ✓ that balances the load on the long arm, so the crane does not tip over ✓.

2.4 A small flat piece (often triangular) fixed over a joint ✓ to make the joint stronger and more rigid ✓. (2)

2.5.1 A clear statement of the problem and what will be designed, e.g. "Design and make a shelter at the taxi rank that protects waiting passengers from sun and rain." ✓✓ (2)

2.5.2 Specifications, any TWO: must keep out rain and sun ✓; seat at least 20 people ✓; strong and stable in wind; safe with no sharp edges; easy to keep clean. (4)
Constraints, any TWO: cost / budget ✓; size of the available space ✓; time to build; materials available; must not block traffic

[18]

QUESTION 3

3.1 (a) Natural ✓ (b) Manufactured ✓ (c) Natural ✓ (d) Manufactured ✓ (4)

3.2 (a) Any TWO: light ✓; waterproof ✓; cheap; can be moulded into any shape; can be transparent; strong (4)
(b) Any TWO: does not rot (not biodegradable) ✓; pollutes land and oceans and harms animals ✓; made from oil (non-renewable); releases toxic gases when burnt

3.3 (a) Any ONE: nails, screws, wood glue, dowels, joints ✓ (b) Any ONE: welding, soldering, rivets, nuts and bolts ✓ (2)

3.4 Any TWO: clamp the work firmly in a vice ✓; wear safety glasses ✓; keep fingers away from the blade; make sure the blade is tight and the teeth point forward; cut slowly with even strokes (2)

[12]

QUESTION 4

4.1 (a) First class ✓ (b) Second class ✓ (c) Third class ✓ (3)

4.2 $(\text{a})\ \text{MA} = 1{,}6 \div 0{,}4\ ✓ = 4\ ✓$ $(\text{b})\ \text{Effort} = 1\,000 \div 4\ ✓ = 250\ \text{N}\ ✓$ (4)

4.3 They give a large, fast movement of the load ✓ and good control (e.g. for picking up small objects) ✓. Example, any ONE: tweezers, fishing rod, broom, tongs, the human forearm ✓ (3)

4.4 (a) Push-pull linkage ✓ (b) Any ONE: bicycle brakes; doorbells; window mechanisms ✓ (2)

4.5 Input: hand squeezes the handles ✓ → Process: the handles turn around the pivot (first-class lever) ✓ → Output: the jaws grip and bend the wire with a greater force ✓ (3)

4.6 Neat sketch of a beam with the fulcrum between the load and the effort ✓; labels E, L and F correct ✓; arrows showing the directions of effort and load ✓ (3)

[18]

QUESTION 5

5.1.1 A battery (two cells) ✓ B switch (closed) ✓ C light bulb ✓ D motor ✓ (4)

5.1.2 The bulb lights up and the motor turns ✓ (1)

5.1.3 No ✓. There is only one path for the current; if the bulb blows, the circuit is broken and no current flows ✓. (2)

5.2 (a) Electrical energy → light energy ✓ and heat energy ✓ (b) Electrical energy → kinetic (movement) energy ✓ (3)

5.3 Copper is a very good conductor of electricity ✓; the plastic is an insulator that protects people from electric shocks and stops short circuits ✓. (2)

5.4 Any TWO: push-button ✓; toggle ✓; slide; rocker; pressure (mat) switch; reed switch; tilt switch (2)

5.5 (a) Two north poles repel (push apart) ✓; north and south poles attract ✓ (b) Iron and steel ✓ (3)

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TOTAL: 75

END OF MARKING GUIDELINE